

Déplacement

Atau Tanaka and Erämaa Trio

Déplacement is a new work for trio (violin, clarinet, piano), muscle sensors and modular synthesizers by **Atau Tanaka** composed for and in partnership with **Erämaa Trio**. It updates the tradition of electroacoustic music and *musique mixte* to create three instrumental subsystems connecting instrumental gesture and performers' bodies to live electronic processing. It uses wearable technologies and concepts of embodied interaction to propose a new work of acoustic/electronic chamber music that we can consider 'post-computer music'.

The work takes the concept of 'déplacement' – movement, mobility, déambulation and *dérive* in shifting reconfigurations of the traditional staging of a classical trio. The work broaches questions of displacement of persons and peoples through history around the world due to forces of culture, religion, economics and politics. Musically, it displaces the acoustic with the electronic and *vice versa*, the body and the instrument, the 'correct' performance position with a foreign place on stage.

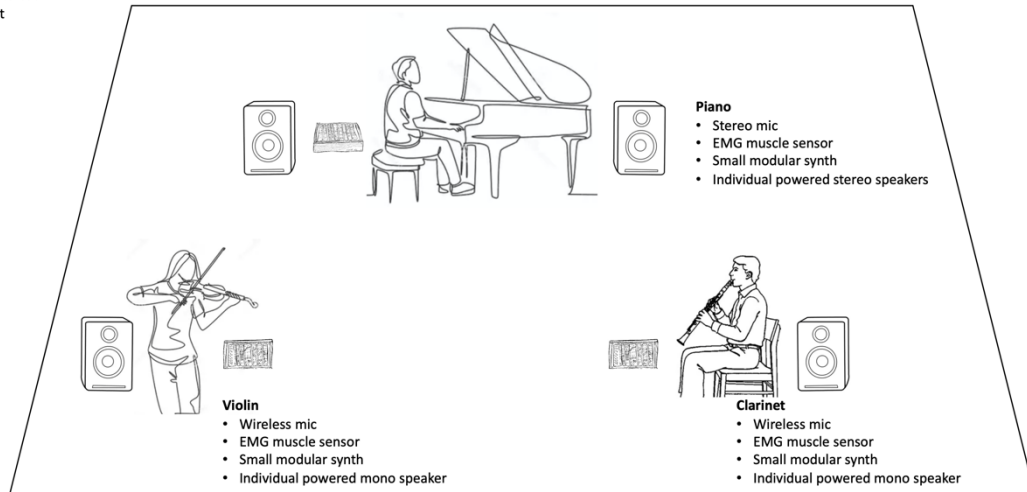
Each of the three musicians in the **Erämaa Trio** will be autonomous in their instrumental technology setup. They will have electromyogram (EMG) electrodes on their body measuring muscle tension and exertion as they play their instrument. The placement of electrodes (arm, shoulder, neck, back) will be determined through experimentation with the performance practice and gestures of each instrument (violin, clarinet, piano). We will use the **EAVI EMG** board, an open-source muscle sensing hardware developed by Tanaka in his research in London at Goldsmiths and in Paris at the MSH Paris Nord. This is the result of research funded by the European Research Council and the French Agence Nationale de Recherche. Each instrument will also have a microphone pickup. The EMG and mic signals will enter a modular synthesizer system consisting of resonators, ring modulars, and granular samplers to produce an electronic sonification of the instrumentalist's body and musical shadow of their instrument. Each member of the trio will have a separate modular synthesizer and speaker system. The electronic sound will be coincident with their position on stage, enabling us to hear the electronic and acoustic sound of each instrument localised together and independently of the others, identifying each soloist musically and spatially. The wireless EMG and mics will give the instrumentalists freedom to move – *de se déplacer sur scène*. This will create a dramaturgy of movement and shifting positions, or musical roles and play. While the localised speaker system allows us to identify each instrument in space, the displacement will blur this space and expand the identity of each performer, enabling forms of call and response with their own shadow and with the other musicians, in proximity and at distance across the stage. The arch of the composition will thus explode the trio from its traditional seating to a displaced, dynamic tension, resolving in a return to home base.

Erämaa Trio commissions Tanaka to compose this new work. Tanaka provides the EAVI EMG boards, and will discuss with Erämaa how to constitute the technical materials. Musically the project will be a developmental process of collaboration where a series of meetings and studio sessions will allow the trio and composer to work together to experiment and become familiar with the muscle sensing and modular treatments. These sessions can be made semi-public in the form of workshops. With the outcomes of these sessions comprising the musical materials, Tanaka will produce a graphical score that is the composition of the work, to then be rehearsed and performed at **Ars Musica 2024**.

Point de départ

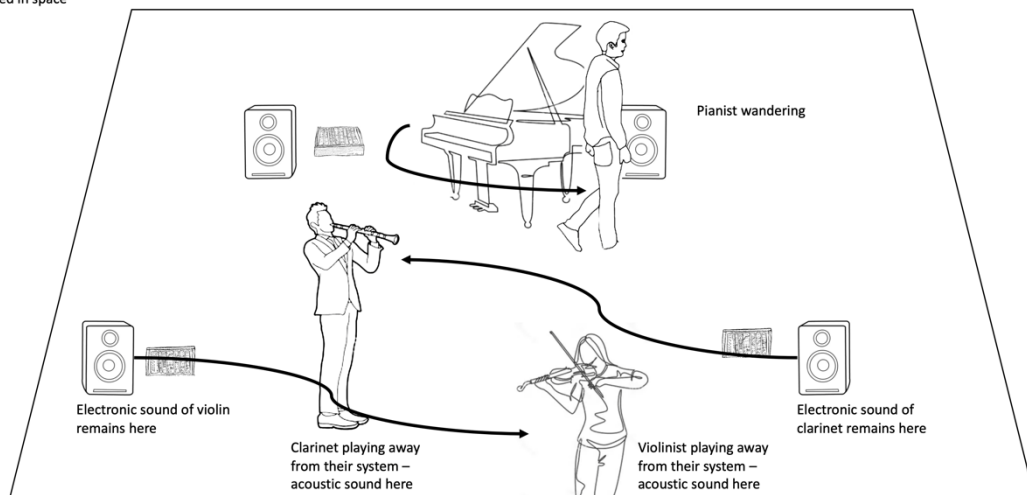
- Performers seated
- Traditional trio staging
- Individual speaker systems to localize sound to each instrumentalist

Atau Tanaka
Erämaa Trio concept
Ars Musica 2024



Trio ambulatoire

- Performers standing
- Wandering the stage
- Acoustic sound and electronic sound separated in space

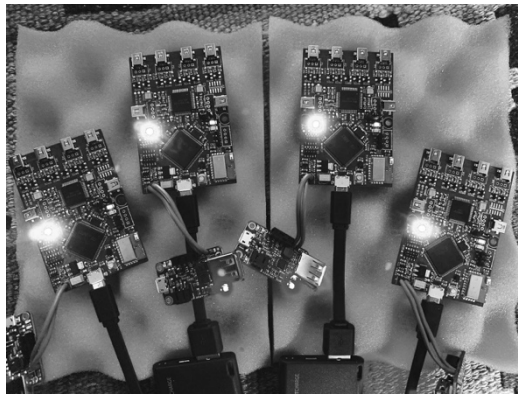


Muscle sensing

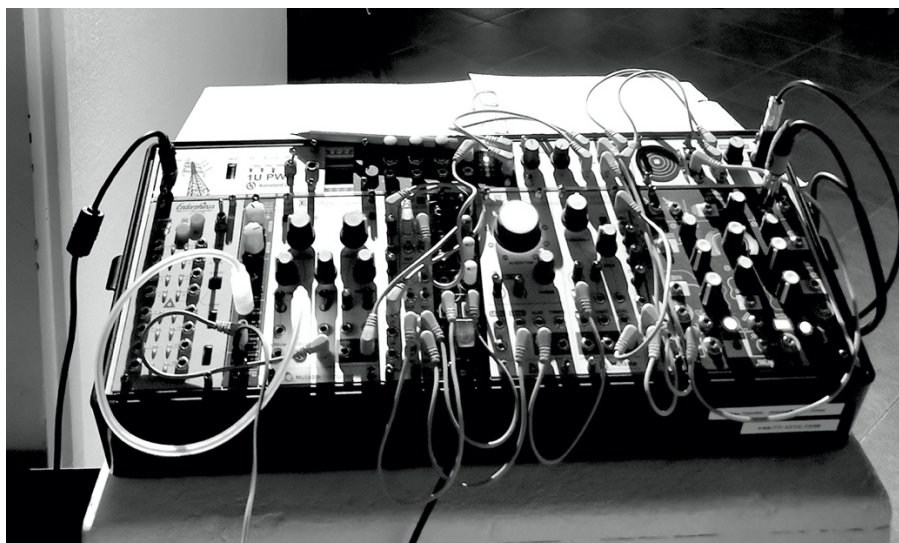
- EMG electrodes
- EAVI wireless interface
- Select muscles best suited for each instrument



Example of cellist with EAVI EMG system



Four EAVI EMG boards



Modular synthesizer